Multimedia Signal Processing - 2nd Module

February 14, 2024

1. **Question (8 pts)**
   Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	$y_1 = 1$	$y_2 = 2$	$y_3 = 3$	$y_4 = 4$
$x_1 = 1$	1/8	0	1/8	0
$x_2 = 2$	0	0	1/4	0
$x_3 = 3$	0	1/4	1/8	0
$x_4 = 4$	0	0	0	1/8

   (a) Compute the probability mass functions for each source separately.
   (b) Compute the entropy of both sources, i.e., $H(X)$ and $H(Y)$.
   (c) Compute the joint entropy $H(X, Y)$.
   (d) Write the two update equations for the Lloyd-Max quantizer.
   (e) Design a 1 bit scalar Lloyd-Max quantizer for the source X only. Initialize the intervals as $I_1 = (-\infty, 2]$ and $I_2 = (2, +\infty]$. Remember that "[" means included, while "(" means excluded, hence $2 \in I_1$.
   (f) How many iterations did you need to reach convergence? If you could not reach it, please comment.
   (g) Compute the expected MSE obtained by quantizing the output of the source X with the defined Lloyd-Max quantizer. (hint: remember how the definition of expected value relates to the concept of mean).

2. **Question (4 pts)** Consider the problem of image coding.
   (a) Report the block diagram of the JPEG image encoder.
   (b) Describe all the blocks in the diagram, clearly stating which is the input and the output of each block (with possible info on the size if needed).
   (c) Explain which blocks take care of visual redundancy and which of visual irrelevancy.

3. **Question (3 pts)** Considering perceptual audio coding.
   (a) Describe the concepts of absolute threshold of hearing.
   (b) Describe the concept of frequency masking.
   (c) Intuitively explain how these concepts can be useful to determine how many bits should be associated to a given tone in perceptual audio coding.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Considering the same distortion level, DPCM coding always provides better compression rate than PCM.
   - [T] [F] $H(X,Y) = H(X) + H(Y)$ for independent sources X and Y.
   - [T] [F] Consider the problem of coding symbols emitted by a source. It is possible to achieve an average bitrate smaller than the entropy of the source.
   - [T] [F] In a video coding scheme, the decoder does not compute motion vectors from the frames.
   - [T] [F] The Huffman code cannot be theoretically worst than the Shannon code.
   - [T] [F] When using a linear predictive vocoder exploiting the source-system model, the coefficients of the synthesis filter are typically directly quantized.
   - [T] [F] In a video coding scheme, the displaced frame difference should in general contain less energy than the simple frame difference if motion estimation and compensation are properly performed.
   - [T] [F] According to rate-distortion theory, the maximum possible distortion after quantization is the source variance (i.e., σ_x^2).

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Extract a single color component (of your choice) from the image.

   (c) Turn the 2D color component in a 1D vector by reading the pixel values by rows or by column.

   DPCM coding

   Considering the 1D color component you extracted from the image, implement a simple DPCM scheme.

   (d) Encode the vector using close-loop DPCM. Use a quantizer of your choice.

   (e) Decode the encoded vector to obtain an estimate of the original 1D signal.

   (f) Compute the approximate bit-rate (in bit per pixel) that you need to store the 1D vector using your DPCM (hint: consider that you are able to apply an ideal lossless coding scheme to the encoded stream).